Robustness in the Face of Partial Identifiability in Reward Learning Problems

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Motivation

Q1 When is partial identifiability *problematic*?
Q2 What is the "best" approach to cope with partial identifiability in Reward Learning (ReL) problems?
Q3 Can we apply it in a provably efficient manner?

Contribution

- Novel **quantitative** framework for ReL
- **Robust** approach to tackle partial identifiability
- Provably efficient robust algorithm for ReL

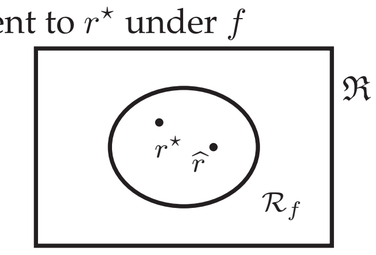
Preliminaries

ReL and Partial Identifiability

- Target reward $r^\star \to$ unknown, to learn
- Feedback $f \to$ gives information on $r^\star$
- Application $g \to$ for which we need $r^\star$
- Partial identifiability $\to f$ doesn't *uniquely* identify $r^\star$

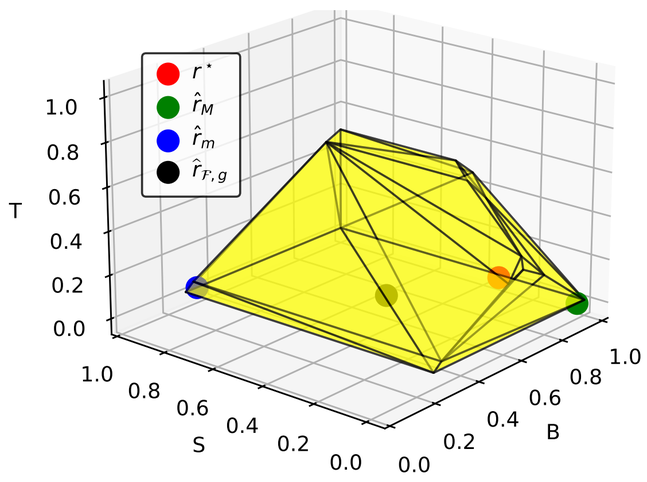
Example

- f = demonstrations $\mathcal{D}^E = \{(s_h, a_h)\} \sim \pi^E$ (IRL)
- f' = preferences $\mathcal{D} = \{(\omega_1, \omega_2, \sigma)\}$ (RLHF)
- g = find an optimal policy under $r^\star$
- $\mathcal{R}_f$ = rewards equivalent to $r^\star$ under f



Illustrative Simulation

- Small MDP with $S = 12, A = 3, H = 5$
- Noisy transition model
- 7 feedback
- State-only 3-dimensional reward
- Error reduction: 1.64 $\to$ 0.82



A New Framework

Ingredients

- Model g as a pair $(\mathcal{X}_g, \mathcal{L}_g)$
- $\mathcal{X}_g \to$ **set** of objects to deploy (e.g., set of policies $\mathcal{X}_g := \Pi$)
- $\mathcal{L}_g : \mathfrak{R} \times \mathcal{X}_g \to \mathbb{R}_+ \to$ **loss** of an object for a reward (e.g., suboptimality $\mathcal{L}_g(r, \pi) := J^*(r) - J^\pi(r)$)

Objective

$$x^\star \in \arg\min_{x \in \mathcal{X}_g} \mathcal{L}_g(r^\star, x)$$

Robust Approach

But $r^\star$ is still unknown $\to$ robust approach!

$$\mathcal{L}_g(r^\star, x) \le \max_{r \in \mathcal{R}_f} \mathcal{L}_g(r, x)$$

Q1 When is partial identifiability problematic?
$\to$ *Quantitative* criterion: if the worst-case error $\mathcal{I}_{f,g} := \min_{x \in \mathcal{X}_g} \max_{r \in \mathcal{R}_f} \mathcal{L}_g(r, x)$ is above some given threshold

Q2 What is the "best" approach to cope with partial identifiability in Reward Learning (ReL) problems?
$\to$ *Robust* approach: deploy the best object in the worst case $x_{f,g} := \arg\min_{x \in \mathcal{X}_g} \max_{r \in \mathcal{R}_f} \mathcal{L}_g(r, x)$

Rob-ReL - A Robust Algorithm

Q3 Can we apply it in a provably efficient manner?
$\to$ Yes in at least the following class of ReL problems

Setting

- Application g: *assess which policy between π^1 and π^2 is preferred under $r^\star$*
  $\to \mathcal{X}_g = \mathbb{R}$, $\mathcal{L}_g(r, x) = \left| x - \left(J^{\pi^1}(r) - J^{\pi^2}(r) \right) \right|$
- Feedback f_1: *demonstrations from (sub-)optimal expert π^E* $\to \mathcal{R}_{f_1} = \left\{ r \in \mathfrak{R} : J^{\pi^E}(r) \ge J^*(r) - \epsilon \right\}$
- Feedback f_2: *hard trajectory comparisons between ω_1 and ω_2* $\to \mathcal{R}_{f_2} = \left\{ r \in \mathfrak{R} : G(\omega_1; r) \ge G(\omega_2; r) \right\}$
- Feedback f_3: *hard policy comparisons between π'_1 and π'_2* $\to \mathcal{R}_{f_3} = \left\{ r \in \mathfrak{R} : J^{\pi'_1}(r) \ge J^{\pi'_2}(r) \right\}$
- Finite data: transition models and policies have to be estimated from online interaction and offline data

Rob-ReL

- Estimate the transition model with **RF-Express** (**?**)
- Estimate the occupancy measure of the policies with the empirical estimates
- Note that $x_{f,g} = (M + m)/$ and $\mathcal{I}_{f,g} = (M - m)/2$, where M, m are max/min over $r \in \mathcal{R}_f$ of $\left(J^{\pi^1}(r) - J^{\pi^2}(r) \right)$
- Solve the optimization problem with the **primal-dual subgradient method** (**?**)

Theorem 1 *In tabular MDPs, under Slater's condition, for suitable choice of hyperparameters,* **`Rob-ReL`** *outputs $\widehat{\mathcal{I}}, \widehat{x}$ such that, with probability $1 - \delta$: $\mathcal{L}_g(r^\star, \widehat{x}) \le \mathcal{I}_{f,g} + \epsilon$ and $|\mathcal{I}_{f,g} - \widehat{\mathcal{I}}| \le \epsilon$, with a number of samples and iterations:*

$$N_\pi \le \widetilde{\mathcal{O}}\left(\frac{SAH^5}{\epsilon^2 \xi^2} \log \frac{1}{\delta} \right), \quad N_p \le \widetilde{\mathcal{O}}\left(\frac{SAH^5}{\epsilon^2 \xi^2} \left(S + \log \frac{1}{\delta} \right) \right), \quad N_{it} \le \widetilde{\mathcal{O}}\left(\frac{(SA)^{3/2} H^5 m}{\epsilon^2 \xi^4} \right).$$